

Supplementary Material: Understanding Software Engineering Organizations Through Systems Thinking

Juliana Alves¹ | Henrique Alves^{1,2} | André Costa
Batista^{1,3}

Nomenclature

| State Variables

F	Product feature completeness
D	Technical debt accumulation
B	Bug count
M	Market share (%)
F_m	Product-market fit
A_{brand}	Brand awareness
A_{org}	Organizational alignment
C_{org}	Organizational conflict
Q_{prod}	Product quality (composite metric)
R	Financial runway (months)
$C_{i,j}$	Capacity j of agent i
V_{agg}	Aggregate development velocity
S_{cap}	Sales capacity
$L_{context}$	Learning context factor
S_{stress}	Stress factor

Parameters - Flow Rates

λ_{feat}	Feature development rate coefficient
λ_{debt}	Technical debt generation rate
λ_{bugs}	Bug introduction rate
λ_{acq}	Customer acquisition rate coefficient
μ_F	Feature deprecation rate
$\rho_{refactor}$	Refactoring effectiveness coefficient
ρ_{fix}	Bug fixing effectiveness coefficient

Parameters - Drag and Quality

δ_{debt}	Technical debt drag coefficient
α_{qual}	Quality investment allocation parameter
α_{HR}	Human Resources coordination capacity (normalized)
κ	Code quality factor
τ	Quality consciousness parameter

Parameters - Financial

γ_{rev}	Revenue conversion coefficient
β_{burn}	Per-employee monthly burn rate
$\beta_{overhead}$	Fixed operational overhead
$F_{funding}(t)$	Funding events at time t

Parameters - Organizational Dynamics

η_{learn}	Base learning rate
η_{decay}	Capability degradation rate
τ_{base}	Baseline monthly turnover probability
ξ_{decay}	Organizational alignment decay rate
$\xi_{conflict}$	Conflict impact coefficient
χ_{base}	Baseline customer churn rate
χ_{bugs}	Bug-driven churn coefficient
χ_{misfit}	Market misfit churn coefficient

Other Notation

n_{total}	Total organizational headcount
i	Agent index
j	Capacity type index
k	Management agent index
h	Human Resources agent index
t	Time (months)

1 | SIMULATION ALGORITHM

Algorithm 1 formalizes the core simulation loop executed each monthly timestep. The algorithm integrates agent capacity evolution, role-specific actions, state updates, strategic decisions, financial dynamics, and team turnover within a unified framework that ensures proper causal ordering.

The algorithm implements seven phases: (1) agent capacity evolution through learning and degradation; (2) role-specific actions producing technical, market, and organizational changes; (3) state integration with feedback dynamics; (4) strategic interventions based on crisis thresholds; (5) financial computation with revenue, burn, and funding; (6) probabilistic turnover and hiring; (7) termination evaluation.

2 | STATE EVOLUTION EQUATIONS

This appendix presents the formal mathematical specifications for state variable evolution. All equations are implemented in discrete-time simulation with monthly timesteps.

2.1 | Technical Subsystem Dynamics

Feature Completeness:

$$\frac{dF}{dt} = \lambda_{feat} \sum_i C_{dev,i} \cdot (1 - \delta_{debt}(D)) - \mu_F \cdot F \quad (1)$$

where the debt drag function implements saturation dynamics:

$$\delta_{debt}(D) = \frac{D}{D + 200} \quad (2)$$

and $\mu_F \approx 0$ (feature depreciation negligible).

Quality Focus Factor: Engineering teams exhibit heterogeneous quality consciousness. To prevent the pathological case where $\overline{C_{exp}} \approx \overline{C_{dev}}$ (leading to $\tau \approx 1$ and $(1 - \tau) \approx 0$, which artificially eliminates debt/bug generation), we normalize quality focus as:

$$\tau = \min \left(\frac{\overline{C_{exp}}}{100}, 0.8 \right) \quad (3)$$

Algorithm 1 Monthly Organizational Simulation Step

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1: Input: State  $S_t$ , Agents  $\mathcal{A}$ , Parameters  $\Theta$ 
2: Output: State  $S_{t+1}$ 
3: for each agent  $a \in \mathcal{A}$  do
4:   Update capacities:  $\Delta C_a \leftarrow \eta_{learn} L_{context}(100 - C_a) - \eta_{decay} S_{stress} C_a$ 
5: end for
6: Compute aggregate outputs by role:
7:   Engineers:  $\Delta F, \Delta D, \Delta B$  from development velocity  $V_{agg}$  with debt drag
8:   Sales:  $\Delta M_{acq}$  from sales capacity, product quality, brand, market saturation
9:   Marketing:  $\Delta A_{brand}$ , lead quality from marketing capacity
10:  HR:  $\Delta A_{org,HR}$ , conflict resolution from coordination capacity
11:  Management:  $\Delta A_{org,mgmt}$  from management capacity
12: Update product & organizational state:  $F, D, B, M, A_{brand}, A_{org}, C_{org}$ 
13: if crisis detected (debt, bugs, or market misfit) then
14:   Adjust strategy: increase  $\alpha_{quality}$  or reallocate effort
15: end if
16: if pivot conditions met then
17:   Execute pivot (reset product, retain team)
18: end if
19: Compute finances:  $Revenue \leftarrow \gamma_{rev} M Q_{prod}$ ;  $Burn \leftarrow \beta_{burn} |\mathcal{A}| + \beta_{overhead}$ 
20: Update runway:  $R \leftarrow R + Revenue - Burn + F_{funding}(R, Q_{prod}, M)$ 
21: for each agent  $a$  do
22:   if  $\text{random}() < P_{turnover}(HR_{eff}, C_{org})$  then
23:     Replace  $a$  with new agent
24:   end if
25: end for
26: if  $R \leq 0$  then return Failure
27: else if  $t > t_{max}$  then return Success
28: end if
29: return  $S_{t+1}$ 

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This ensures $(1 - \tau) \in [0.2, 1.0]$, creating debt accumulation dynamics even when teams have high explanation capacity.

Technical Debt Evolution:

$$\frac{dD}{dt} = \lambda_{debt} \cdot V_{agg} \cdot (1 - \tau) - \rho_{refactor} \cdot W_{refactor} \quad (4)$$

where V_{agg} is aggregate development velocity, $W_{refactor}$ is refactoring work allocation, and $\rho_{refactor} = 0.04$ represents refactoring effectiveness.

Exponential Velocity Penalty: Technical debt imposes a non-linear drag on development velocity:

$$\delta_{debt}(D) = \frac{1}{1 + (D/50)^2} \quad (5)$$

This exponential penalty ensures that accumulated debt ($D > 100$) drastically reduces productivity, creating the feedback dynamics of Loop R1. When $D > 200$, the system enters technical bankruptcy (game over condition).

Code Quality Evolution:

$$\frac{dQ_c}{dt} = \gamma_{refactor} \cdot W_{refactor} - \gamma_{decay} \cdot D \quad (6)$$

with $\gamma_{refactor} = 0.01$ and $\gamma_{decay} = 0.01$, implementing bidirectional coupling between quality and debt.

Bug Count Evolution:

$$\frac{dB}{dt} = \lambda_{bugs} \cdot V_{agg} \cdot \kappa(Q_c) \cdot (1 - \tau) - \rho_{fix} \cdot W_{bugfix} \quad (7)$$

where the quality modulation function is:

$$\kappa(Q_c) = \frac{100 - Q_c}{100} \quad (8)$$

and ρ_{fix} represents bug fixing efficiency.

Bug Velocity Penalty: Bugs impose exponential drag on development velocity:

$$\delta_{bugs}(B) = \frac{1}{1 + (B/20)^{1.5}} \quad (9)$$

Beyond critical thresholds ($B > 100$), the system becomes unmaintainable, triggering game over.

2.2 | Resource Allocation Strategy

Engineering capacity is allocated according to strategic parameters α_{dev} and $\alpha_{features}$:

$$W_{dev} = C_{available} \cdot \alpha_{dev} \quad (10)$$

$$W_{quality} = C_{available} \cdot (1 - \alpha_{dev}) \quad (11)$$

$$W_{features} = W_{dev} \cdot \alpha_{features} \quad (12)$$

$$W_{refactor} = W_{dev} \cdot (1 - \alpha_{features}) \quad (13)$$

$$W_{bugfix} = W_{quality} \quad (14)$$

where $C_{available}$ is total available engineering capacity.

2.3 | Market Subsystem Dynamics

Product Quality Composite:

$$Q_{prod} = \frac{F_m}{100} \cdot \frac{F}{F + 50} \cdot \left(1 - \frac{B}{B + 50}\right) \quad (15)$$

combining market fit, feature completeness (with saturation), and bug-free operation.

Lead Quality:

$$L_q = \text{clip} \left(\frac{\overline{C_{pub}} + \overline{C_{ideas}}}{2} \cdot Q_{prod}, 0, 100 \right) \quad (16)$$

where $\overline{C_{pub}}$ and $\overline{C_{ideas}}$ are mean marketing capacities.

Brand Awareness Growth:

$$\frac{dA_{brand}}{dt} = \omega_{mkt} \cdot M_{eff} \cdot \left(1 - \frac{A_{brand}}{100}\right) - \mu_{brand} \cdot A_{brand} \quad (17)$$

where M_{eff} is effective marketing effort and decay μ_{brand} represents brand erosion.

Market Share Dynamics:

$$\frac{dM}{dt} = \lambda_{acq} \cdot S_{cap} \cdot Q_{prod} \cdot A_{brand} \cdot (1 - M/100)^2 - \chi_{total} \cdot M \quad (18)$$

where total churn is:

$$\chi_{total} = \chi_{base} + \chi_{bugs} \cdot \frac{B}{B + 50} + \chi_{misfit} \cdot \left(1 - \frac{F_m}{100}\right) \quad (19)$$

implementing multi-factor attrition.

2.4 | Organizational Subsystem Dynamics

Organizational Alignment:

$$\frac{dA_{org}}{dt} = \sum_k C_{mgmt,k} + \sum_h C_{coord,h} - \xi_{decay} \cdot A_{org} - \xi_{conflict} \cdot C_{org} \quad (20)$$

with half-life approximately 15 months without active management ($\xi_{decay} \approx 0.047$).

Conflict Generation:

$$\frac{dC_{org}}{dt} = \gamma_{misalign} \cdot \left(1 - \frac{A_{org}}{100}\right) + \gamma_{pressure} \cdot \frac{B}{B+50} - \rho_{HR} \cdot \sum_h C_{resolve,h} \quad (21)$$

where $\gamma_{misalign} = 0.3$ and $\gamma_{pressure} = 0.2$ represent conflict generation rates, and ρ_{HR} is HR resolution effectiveness.

2.5 | Financial Dynamics

Revenue Generation:

$$R_{revenue} = \gamma_{rev} \cdot Q_{prod} \cdot S_{power} \cdot M \quad (22)$$

where sales power is:

$$S_{power} = \overline{C_{sell}} \cdot n_{sales} \quad (23)$$

with $\overline{C_{sell}}$ mean selling capacity and n_{sales} sales team size.

Cash Runway Evolution:

$$\frac{dR_{cash}}{dt} = R_{revenue} + F_{funding}(t) - (\beta_{burn} \cdot n_{total} + \beta_{overhead}) \quad (24)$$

where runway depletion ($R_{cash} \leq 0$) triggers crisis protocols or venture termination.

2.6 | Quality Penalties on Revenue

Product quality defects directly impact revenue generation through exponential penalty functions:

$$\text{bug_penalty} = \frac{1}{1 + (B/30)^2} \quad (25)$$

$$\text{debt_penalty} = \frac{1}{1 + (D/80)^{1.5}} \quad (26)$$

These penalties multiply with $R_{revenue}$, ensuring that technical shortcuts eventually erode market performance.

2.7 | Game Over Conditions

The model implements three terminal conditions:

1. Financial Bankruptcy: $R_{cash} \leq 0$ (cash runway exhausted)
2. Technical Bankruptcy: $D > 200$ (technical debt exceeds maintainability threshold)
3. Quality Crisis: $B > 100$ (bug count exceeds customer tolerance)

These conditions formalize the multi-dimensional failure modes observed in software ventures.

2.8 | Capacity Evolution Mechanisms

Agent capabilities evolve through learning and degradation processes:

$$\Delta C_{i,j} = \eta_{learn} \cdot L_{context} \cdot (100 - C_{i,j}) - \eta_{decay} \cdot S_{stress} \cdot C_{i,j} \quad (27)$$

where learning context incorporates:

$$L_{context} = L_{base} \cdot \left(1 + \frac{A_{org}}{100}\right) \cdot \left(1 + \frac{TD_{boost}}{100}\right) \cdot F_{focus} \quad (28)$$

with TD_{boost} representing HR talent development contribution and F_{focus} strategic focus alignment.

Stress factors include:

$$S_{stress} = S_{base} \cdot \left(1 + \frac{C_{org}}{100}\right) \cdot \left(1 + \frac{B}{B + 50}\right) \quad (29)$$

implementing degradation under conflict and quality crises.

Turnover Dynamics:

$$P_{turnover} = \tau_{base} \cdot \left(1.5 - \frac{\overline{C_{HR}}}{100}\right) \cdot \left(1 + 0.5 \cdot \frac{C_{org}}{100}\right) \quad (30)$$

where $\tau_{base} = 0.02$ (2% monthly baseline) and $\overline{C_{HR}}$ represents normalized HR coordination capacity.

3 | STATE VARIABLE SPECIFICATIONS

This section provides detailed mathematical specifications for each of the six primary state variables and their governing equations.

3.1 | Product Features (F)

Accumulated functionality representing product scope. Growth is non-linear due to technical debt's velocity penalty ($\delta_{debt} \cdot D$), creating path-dependent development trajectories where early shortcuts compound into long-term velocity loss.

Equation:

$$\frac{dF}{dt} = \lambda_{feat} \sum_i C_{dev,i} \cdot (1 - \delta_{debt} \cdot D) - \mu_F \cdot F \quad (31)$$

where $\mu_F \approx 0$ (negligible depreciation), λ_{feat} is the feature development rate coefficient, $C_{dev,i}$ is development capacity of engineer i , and δ_{debt} is the technical debt drag coefficient.

3.2 | Technical Debt (D)

Accumulated design compromises and shortcuts. Represents debt principal (D), while the debt drag coefficient (δ_{debt}) operationalizes interest as an ongoing productivity tax on feature development.

Equation:

$$\frac{dD}{dt} = \lambda_{debt} \cdot V_{agg} \cdot (1 - \tau) - \rho_{refactor} \cdot W_{refactor} \quad (32)$$

with exponential penalty $\delta_{debt}(D) = 1/(1 + (D/50)^2)$, implementing Loops R1/B1. Here λ_{debt} is the technical debt generation rate, V_{agg} is aggregate development velocity, τ is quality consciousness parameter, $\rho_{refactor}$ is refactoring effectiveness coefficient, and $W_{refactor}$ is refactoring work allocation.

3.3 | Bug Count (B)

Known defects awaiting resolution. Exhibits threshold behavior—beyond critical levels ($B > 30$), bugs cascade through system interdependencies faster than resolution capacity, triggering Loop R2 (Bug Accumulation) if not arrested by crisis management.

Equation:

$$\frac{dB}{dt} = \lambda_{bugs} \cdot V_{agg} \cdot \kappa \cdot (1 - \tau) - \rho_{fix} \cdot W_{bugfix} \quad (33)$$

with penalty $\delta_{bugs}(B) = 1/(1 + (B/20)^{1.5})$, implementing Loop R2. Here λ_{bugs} is the bug introduction rate, κ is code quality factor, ρ_{fix} is bug fixing effectiveness coefficient, and W_{bugfix} is bug fixing work allocation.

3.4 | Market Share (M)

Captured market percentage ($M \in [0, 100]$). Saturation effects create diminishing returns: $(1 - M)^2$ models increasing difficulty of capturing remaining market. Multi-factor churn (base, bug-driven, misfit-driven) implements Loop B2 (Market Feedback). **Equation:**

$$\frac{dM}{dt} = \lambda_{acq} S_{cap} Q_{prod} A_{brand} (1 - M)^2 - (\chi_{base} + \chi_{bugs} B + \chi_{misfit} (100 - F_m)) M \quad (34)$$

where λ_{acq} is customer acquisition rate coefficient, S_{cap} is sales capacity, Q_{prod} is product quality, A_{brand} is brand awareness, χ_{base} is baseline customer churn rate, χ_{bugs} is bug-driven churn coefficient, χ_{misfit} is market misfit churn coefficient, and F_m is product-market fit. The saturation constraint $(1 - M)^2$ implements Loops R3/B2.

3.5 | Organizational Alignment (A_{org})

Degree of strategic coherence across teams ($A_{org} \in [0, 100]$). Acts as a capacity multiplier—misaligned organizations waste effort on internal friction rather than value creation. Natural entropy (ξ_{decay}) requires continuous management and HR coordination to sustain, implementing Loop B4 dynamics.

Equation:

$$\frac{dA_{org}}{dt} = \sum_k C_{mgmt,k} + \sum_h C_{coord,h} - (\xi_{decay} A_{org} + \xi_{conflict} C_{org}) \quad (35)$$

where $C_{mgmt,k}$ is management capacity of manager k , $C_{coord,h}$ is coordination capacity of HR agent h , ξ_{decay} is organizational alignment decay rate (half-life ≈ 15 months without active management), $\xi_{conflict}$ is conflict impact coefficient, and C_{org} is organizational conflict level.

3.6 | Financial Runway (R)

Available capital measured in months of operation (R). Revenue inflow depends on market share and product quality; burn rate scales linearly with headcount plus fixed overhead. Depletion triggers crisis responses (layoffs, pivots) or venture failure, implementing critical constraint on organizational dynamics.

Equation:

$$\frac{dR}{dt} = \gamma_{rev} M Q_{prod} + F_{funding}(t) - (\beta_{burn} n_{total} + \beta_{overhead}) \quad (36)$$

where γ_{rev} is revenue conversion coefficient, $F_{funding}(t)$ represents funding events at time t , β_{burn} is per-employee monthly burn rate, n_{total} is total organizational headcount, and $\beta_{overhead}$ is fixed operational overhead. Depletion ($R \leq 0$) triggers game over or crisis response.

4 | ADDITIONAL FIGURES

This section contains supplementary figures from the computational experiments.

4.1 | Experiment 1: Unbalanced Team Configurations

4.2 | Experiment 2: Quality vs. Speed

4.3 | Experiment 3: Initial Capital

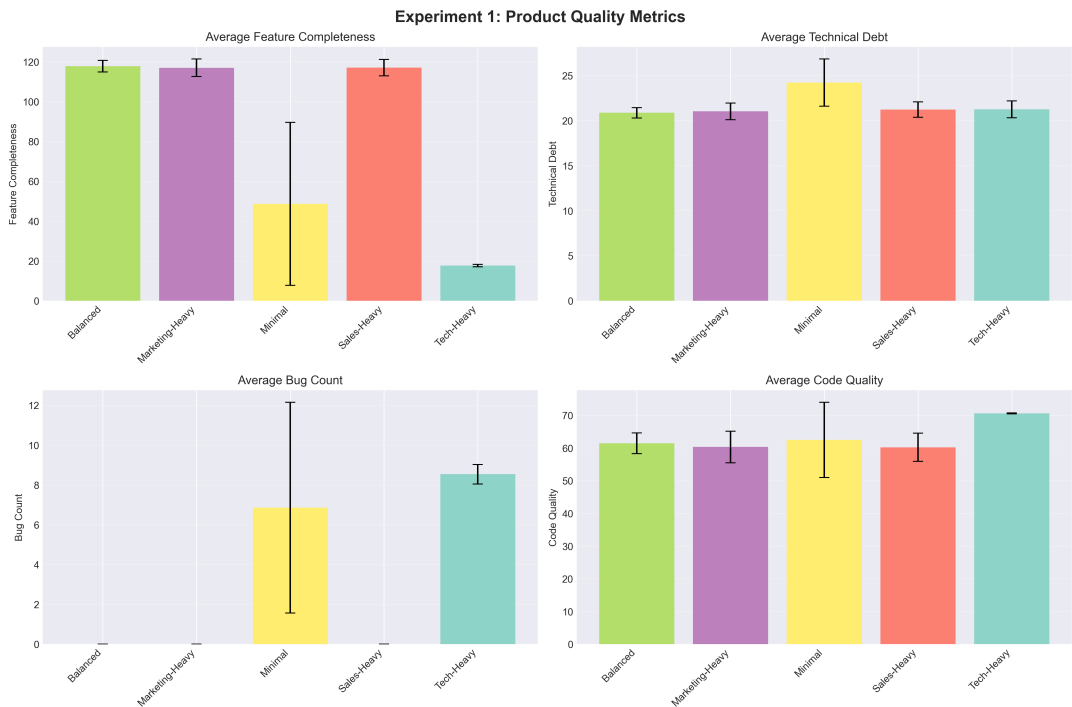


FIGURE 1 Experiment 1: Product quality metrics reveal failure-driven degradation. Top-left: Features stratify. Top-right: Debt bifurcates. Bottom-left: Bugs separate failures from survivors. Bottom-right: Code quality clusters.

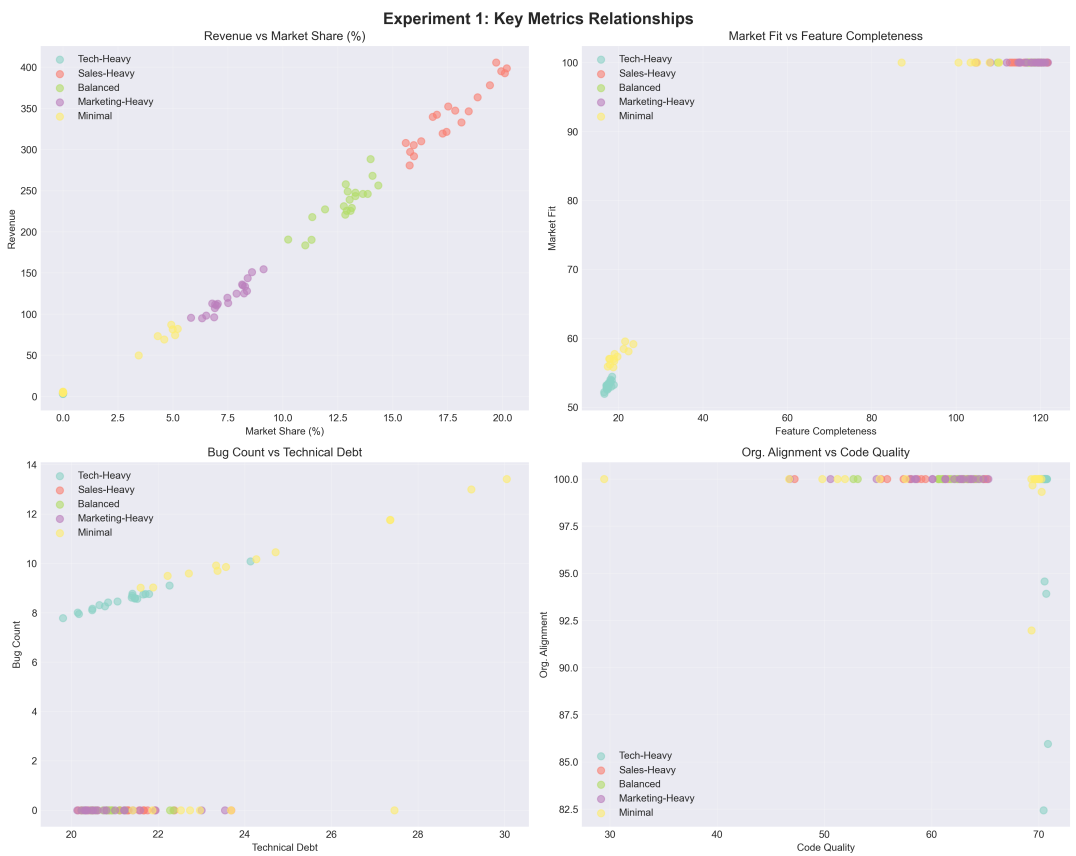


FIGURE 2 Experiment 1: Comparative metrics across configurations. Top-left: Revenue vs. market share. Top-right: Market fit reaches ceiling (100%) for all survivors. Bottom-left: Technical debt and bugs remain zero for all survivors. Bottom-right: Organizational alignment.



FIGURE 3 Experiment 1: Organizational health metrics show minimal differentiation. Left: Organizational alignment remains uniformly high. Right: Organizational conflict remains precisely zero for all configurations.

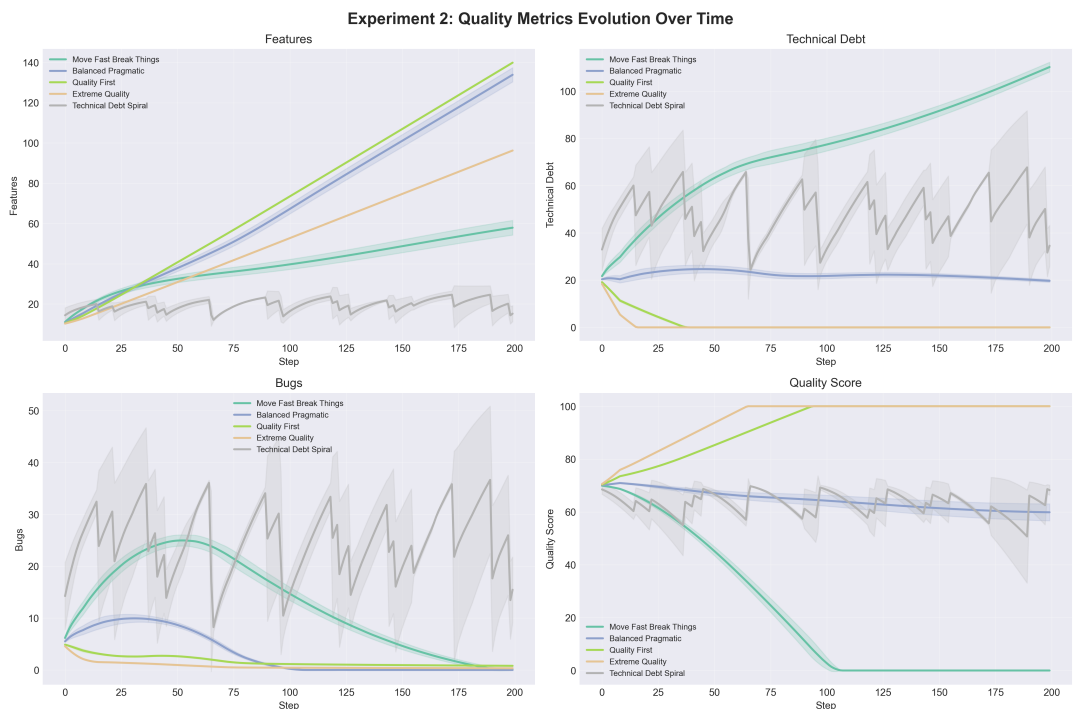


FIGURE 4 Experiment 2: Quality metrics evolution demonstrates functional quality degradation mechanisms. Top-left: Features stratify by velocity. Top-right: Technical debt. Bottom-left: Bug count. Bottom-right: Code quality.

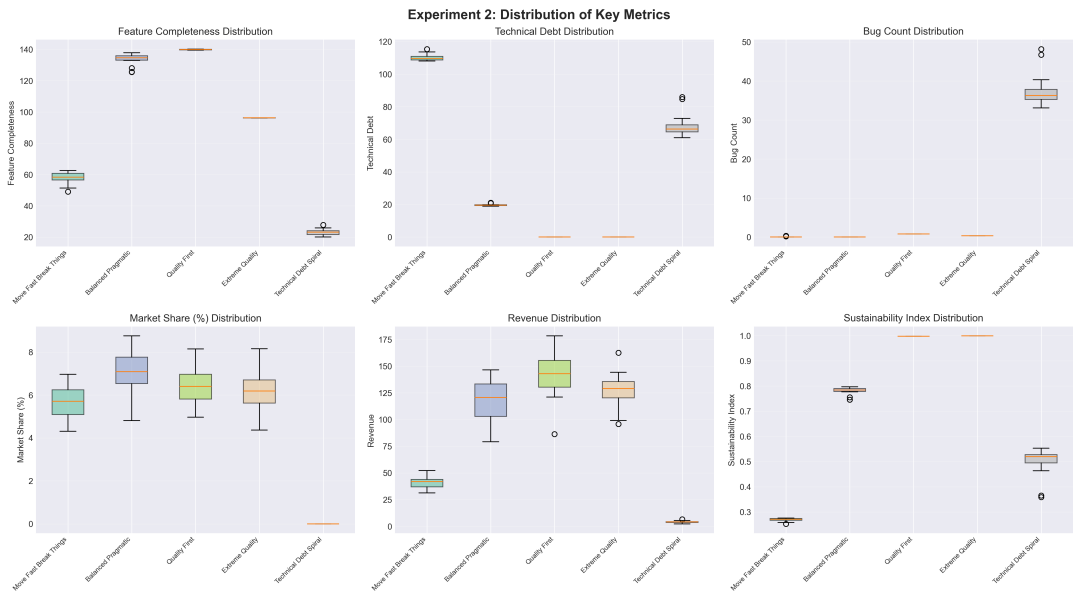


FIGURE 5 Experiment 2: Performance distributions reveal quality-velocity tradeoff emergence.

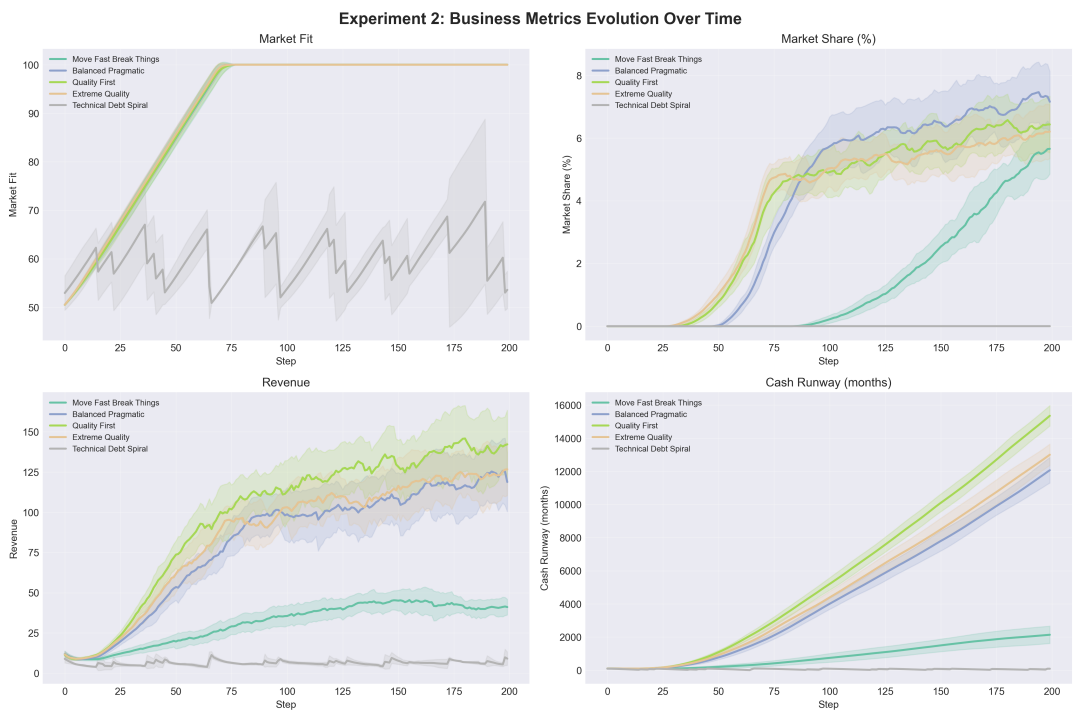


FIGURE 6 Experiment 2: Business metrics evolution shows quality-driven bifurcation.

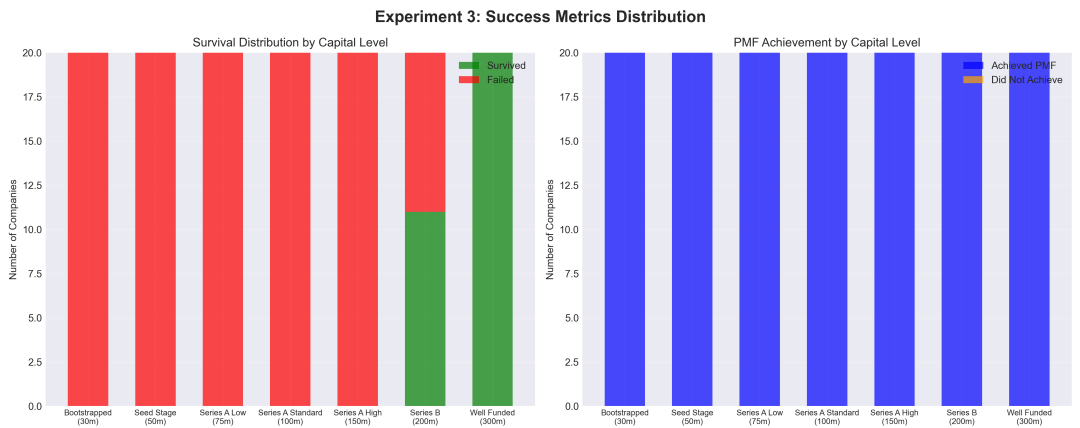


FIGURE 7 Experiment 3: Metric distributions by capital level.

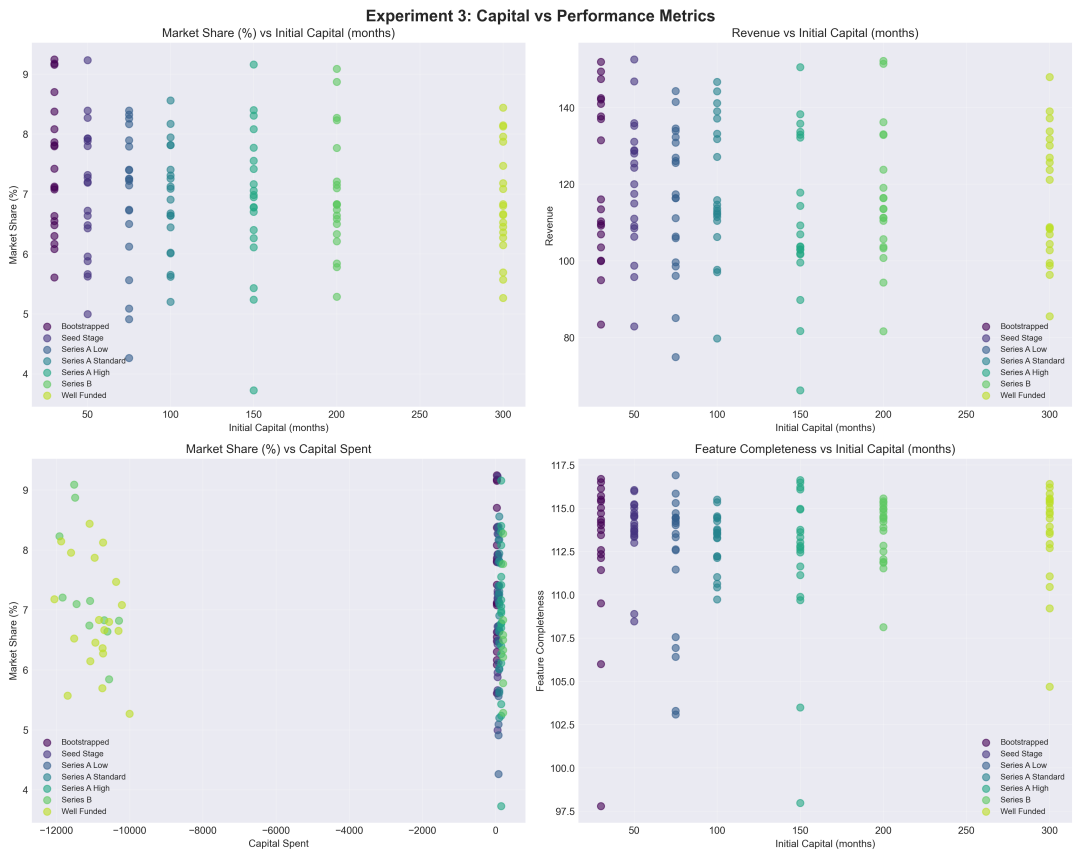


FIGURE 8 Experiment 3: Capital versus performance metrics.